Tushar Kanti Dey

Full-Stack Software Engineer — Scalable Systems Builder

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Professional Summary

Computer Science undergraduate with strong foundations in data structures, algorithms, and backend system design. Experienced in architecting full-stack applications with structured data models, secure authentication flows, and modular codebases. Focused on performance optimization, scalability, and production-grade engineering practices.

Core Competencies

Frontend Engineering Next.js, React.js, Tailwind CSS, TypeScript, Component Architecture

Backend Engineering Node.js, NestJS, Express.js, RESTful API Design, Authentication Systems

Data Systems PostgreSQL, Supabase, MongoDB, Redis, Schema Design

Computer Science Foundations Data Structures, Algorithms, OOP, DBMS, Operating Systems

Mobile Systems React Native, Expo, Rendering Optimization

Engineering Tooling Git, GitHub, Docker, CI/CD, Vercel

Experience

Jun 2024 – Present

Full Stack Engineer, *AZMTH*

Impact: Delivered production-grade applications with improved reliability, performance, and engineering velocity.

Key Contributions:

- Reduced frontend development time by **30%+** by building reusable component and service patterns across modules.
- Improved deployment stability, reducing runtime failures by **40%** through CI/CD pipelines and environment validation.
- Designed scalable APIs and schema structures supporting concurrent usage and efficient data flow.
- Monitored runtime systems and resolved production issues, improving uptime consistency.

Technologies: Next.js, Tailwind CSS, Supabase, Git, Vercel

Feb 2025 – Present

Co-Tech Lead, *CYCODERS*

Impact: Led engineering execution for a scalable CS club platform with structured architecture and maintainable systems.

Key Contributions:

- Designed backend contracts and data models, reducing future feature complexity by **35%**.
- Delivered full-stack features independently, from schema design to deployment.
- Mentored **5+ contributors**, improving code quality and reducing review cycles.
- Structured modular codebases improving maintainability and onboarding speed.

Technologies: TypeScript, Node.js, Supabase, Git, Vercel

Mar 2025 – Present **Designer & Developer**, *Namespace*
Impact: Built high-quality UI systems with strong design-engineering alignment and optimized performance.
Key Contributions:
- Reduced UI inconsistencies by **40%+** through reusable component libraries and design tokens.
- Converted Figma designs into production-ready TypeScript implementations with minimal rework.
- Improved design-to-engineering collaboration through structured implementation workflows.
- Optimized frontend performance across multiple device classes.

Technologies: Figma, GIMP, Inkscape, TypeScript, Supabase, Docker, Google Cloud

Signature Projects

WebScope – Data Extraction and Processing System
Overview: Designed a backend-driven web data extraction platform focused on structured parsing and API delivery.
Engineering Contributions:
- Built modular scraping pipelines handling structured and semi-structured data sources.
- Designed REST APIs with validation and error isolation for reliable data delivery.
- Improved processing efficiency through structured transformation layers.
- Maintained clean architecture using separation of concerns.

Tech Stack: Next.js, Node.js, REST APIs.

Fenix – Real-Time Communication Platform
Overview: Architected a real-time video communication platform using WebRTC infrastructure.
Engineering Contributions:
- Integrated LiveKit for low-latency video and audio streaming.
- Implemented authenticated sessions and secure meeting lifecycle flows.
- Designed scalable UI architecture for predictable rendering.
- Ensured stable real-time communication under varying network conditions.

Tech Stack: Next.js, LiveKit, Clerk, Tailwind CSS.

Signifiya – Performance-Optimized Mobile Application
Overview: Built a mobile application focused on runtime performance and efficient API integration.
Engineering Contributions:
- Reduced unnecessary re-renders improving UI responsiveness.
- Optimized application size through media compression and config tuning.
- Implemented secure environment-based API configurations.
- Managed Android build and deployment lifecycle.

Tech Stack: React Native, Expo, REST APIs.

Taskhub – Modular Task Management System
Overview: Developed a modular task tracking system with structured workflows.
Engineering Contributions:
- Designed predictable state management for task lifecycle.
- Built reusable UI components for maintainability.
- Ensured responsive and device-agnostic layout.
- Deployed optimized production build via Vercel.

Tech Stack: Next.js, React, Tailwind CSS.

Education

2023–Present **B.Tech in Computer Science Engineering**, *Adamas University*, Kolkata, India

Key Strengths

- Strong grasp of algorithmic complexity and structured problem solving.
- Experience designing scalable and maintainable system architectures.
- Clear understanding of database schema design and API contracts.
- Proficient in debugging, optimizing, and deploying production systems.